

HDI Predictor Documentation

Data Flow Diagram

Date	12 June 2025
Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor - Human Development Index Classification System
Prepared By	Kankipati Sai Venkat

DFD Symbol Legend

Symbol	Name	Description
Oval	External Entity	A person or system outside the project boundary (e.g., the User)
Rectangle (numbered)	Process	An activity that transforms incoming data into outgoing data
Rectangle (labelled D)	Data Store	A place where data is held for later use (e.g., the trained model file)
Labelled arrow	Data Flow	The movement of data between entities, processes, and stores

Level-1 Data Flow Diagram



Data Flow Diagram

Flow narrative: 1. The **User** submits four core indicators — life expectancy, mean years of schooling, expected years of schooling, and GNI per capita (PPP) — through the Flask web form. 2. **Process 1.0 (Validate Input)** checks that all fields are present and numeric before passing them downstream. 3. **Process 2.0 (Preprocess & Scale Features)** applies the same scaling/encoding used during training, pulling reference parameters from the training dataset (**D2**). 4. **Process 3.0 (RandomForest Classifier)** loads the serialized model (**D1 - model.pkl**) and produces a predicted HDI tier along with class probabilities. 5. **Process 4.0 (Render Result)** formats the tier, numeric HDI score, and a short feature summary, and returns it to the User.